A cheap version of Halasz's inequality

Terence Tao · 23 November 2015

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A basic estimate in multiplicative number theory (particularly if one is using the Granville-Soundararajan "pretentious" approach to this subject) is the following inequality of Halasz (formulated here in a quantitative form introduced by Montgomery and Tenenbaum).

> **Theorem 1 (Halasz inequality)** Let $f : \mathbf{N} \to \mathbf{C}$ be a multiplicative function bounded in magnitude by 1, and suppose that $x \geq 3$, $T \geq 1$, and $M \geq 0$ are such that
>
> $$\sum_{p \leq x} \frac{1 - \mathrm{Re}(f(p)p^{-it})}{p} \geq M \tag{1}$$
>
> for all real numbers t with $|t| \leq T$. Then
>
> $$\frac{1}{x} \sum_{n \leq x} f(n) \ll (1+M)e^{-M} + \frac{1}{\sqrt{T}}.$$

As a qualitative corollary, we conclude (by standard compactness arguments) that if

$$\sum_{p} \frac{1 - \mathrm{Re}(f(p)p^{-it})}{p} = +\infty$$

for all real t, then

$$\frac{1}{x} \sum_{n \leq x} f(n) = o(1) \tag{2}$$

as $x \to \infty$. In the more recent work of this paper of Granville and Soundararajan, the sharper bound

$$\frac{1}{x} \sum_{n \leq x} f(n) \ll (1+M)e^{-M} + \frac{1}{T} + \frac{\log\log x}{\log x}$$

is obtained (with a more precise description of the $(1+M)e^{-M}$ term).

The usual proofs of Halasz's theorem are somewhat lengthy (though there has been a recent simplification, in forthcoming work of Granville, Harper, and Soundarajan). Below the fold I would like to give a relatively short proof of the following "cheap" version of the inequality, which has slightly weaker quantitative bounds, but still suffices to give qualitative conclusions such as (2).

> **Theorem 2 (Cheap Halasz inequality)** Let $f : \mathbf{N} \to \mathbf{C}$ be a multiplicative function bounded in magnitude by 1. Let $T \geq 1$ and $M \geq 0$, and suppose that x is sufficiently large depending on T, M. If (1) holds for all $|t| \leq T$, then

$$\frac{1}{x}\sum_{n\leq x} f(n) \ll (1+M)^{1/6} e^{-M/3} + \frac{1}{T}.$$

The non-optimal exponent $1/3$ can probably be improved a bit by being more careful with the exponents, but I did not try to optimise it here. A similar bound appears in the first paper of Halasz on this topic.

The idea of the argument is to split f as a Dirichlet convolution $f = f_1 * f_2 * f_3$ where f_1, f_2, f_3 is the portion of f coming from "small", "medium", and "large" primes respectively (with the dividing line between the three types of primes being given by various powers of x). Using a Perron-type formula, one can express this convolution in terms of the product of the Dirichlet series of f_1, f_2, f_3 respectively at various complex numbers $1+it$ with $|t| \leq T$. One can use L^2 based estimates to control the Dirichlet series of f_2, f_3, while using the hypothesis (1) one can get L^∞ estimates on the Dirichlet series of f_1. (This is similar to the Fourier-analytic approach to ternary additive problems, such as Vinogradov's theorem on representing large odd numbers as the sum of three primes.) This idea was inspired by a similar device used in the work of Granville, Harper, and Soundarajan. A variant of this argument also appears in unpublished work of Adam Harper.

I thank Andrew Granville for helpful comments which led to significant simplifications of the argument.

— 1. Basic estimates —

We need the following basic tools from analytic number theory. We begin with a variant of the classical Perron formula.

Proposition 3 (Perron type formula) Let $f : \mathbf{N} \to \mathbf{C}$ be an arithmetic function bounded in magnitude by 1, and let $x, T \geq 1$. Assume that the Dirichlet series $F(s) := \sum_{n=1}^\infty \frac{f(n)}{n^s}$ is absolutely convergent for $\mathrm{Re}(s) \geq 1$. Then

$$\frac{1}{x}\sum_{n\leq x} f(n) \ll \int_{-T}^T |F(1+it)| \frac{dt}{1+|t|} + \frac{1}{T} + \frac{T}{x}.$$

Proof: By telescoping series (and treating the contribution of $n \ll T$ trivially), it suffices to show that

$$\sum_{x/e\leq n\leq x} f(n) \ll x \int_{-T}^T |F(1+it)| \frac{dt}{1+|t|} + \frac{x}{T}$$

whenever $x \geq T$.

The left-hand side can be written as

$$x \sum_n \frac{f(n)}{n} g(\log x - \log n) \tag{3}$$

where $g(u) := e^{-u} 1_{[0,1]}(u)$. We now introduce the mollified version

$$\tilde{g}(u) := \int_{\mathbf{R}} g(u + \frac{v}{T})\Psi(v)\ dv$$

of g, where

$$\Psi(v) := \frac{1}{2\pi} \int_{\mathbf{R}} \psi(t) e^{itv}\ dt$$

and $\psi : \mathbf{R} \to \mathbf{R}$ is a fixed smooth function supported on $[-1,1]$ that equals 1 at the origin. Basic Fourier analysis then tells us that Ψ is a Schwartz function with total mass one. This gives the crude bound

$$\tilde{g}(u) \ll 1$$

for any u. For $u \leq -\frac{1}{T}$ or $u \geq 1 + \frac{1}{T}$, we use the bound $\Psi(v) \ll \frac{1}{(1+|v|)^{10}}$ (say) to arrive at the bound

$$\tilde{g}(u) \ll (T\ \mathrm{dist}(u, \{0,1\}))^{-9};$$

for $\frac{1}{T} \leq u \leq 1 - \frac{1}{T}$ we again use $\Psi(v) \ll \frac{1}{(1+|v|)^{10}}$ and write

$$\tilde{g}(u) - g(u) = \int_{\mathbf{R}} (g(u + \frac{v}{T}) - g(u))\Psi(v)\ dv$$

and use the Lipschitz bound $g(u + \frac{v}{T}) - g(u) \ll \frac{v}{T}$ for $u + \frac{v}{T} \in [0,1]$ to obtain

$$\tilde{g}(u) - g(u) \ll (T\ \mathrm{dist}(u, \{0,1\}))^{-9}$$

for such u. Putting all these bounds together, we see that

$$\tilde{g}(u) = g(u) + O(\frac{1}{(1+T|u|)^9}) + O(\frac{1}{(1+T|u-1|)^9})$$

for all u. In particular, we can write (3) as

$$x \sum_n \frac{f(n)}{n} \tilde{g}(\log x - \log n)$$

$$+O(\sum_n \frac{x}{n} \frac{1}{(1+T|\log x - \log n|)^9} + \frac{1}{(1+T|\log x/e - \log n|)^9}).$$

The expression $\frac{x}{n}\frac{1}{(1+T|\log x - \log n|)^9} + \frac{1}{(1+T|\log x/e - \log n|)^9}$ is bounded by $O(\frac{x}{n}\frac{1}{T\log^9 x})$ when $n \leq x/10$, is bounded by $O(\frac{x}{n}\frac{1}{T\log^9 n})$ when $n \geq 10x$, is bounded by $O(1)$ when $n = (1+O(1/T))x$ or $n = (1+O(1/T))x/e$, and is bounded by $(\frac{T}{x}|n-x|)^{-9} + (\frac{T}{x}|n - x/e|)^{-9}$ otherwise. From these bounds, a routine calculation (using the hypothesis $x \geq T$) shows that

$$\sum_n \frac{x}{n} \frac{1}{(1+T|\log x - \log n|)^9} + \frac{1}{(1+T|\log x/e - \log n|)^9} \ll \frac{x}{T}$$

and so it remains to show that

$$\sum_n \frac{f(n)}{n} \tilde{g}(\log x - \log n) \ll \int_{-T}^{T} |\ F(1+it)|\ \frac{dt}{1+|t|}.$$

Writing

$$\begin{aligned}\tilde{g}(u) &= T \int_{\mathbf{R}} g(v) \Psi(T(v-u))\ dv \\ &= \frac{T}{2\pi} \int_{\mathbf{R}} \psi(t) G(Tt) e^{-iTtu} dt \\ &= \frac{1}{2\pi} \int_{\mathbf{R}} \psi(t/T) G(t) e^{-itu}\ dt\end{aligned}$$

where

$$G(t) := \int_{\mathbf{R}} g(v) e^{itv}\ dv$$

we see from the triangle inequality and the support of $\psi(t/T)$ that

$$\begin{aligned}\sum_n \frac{f(n)}{n} \tilde{g}(\log x - \log n) &\ll \int_{|t|\leq T} |G(t)| \left| \sum_n \frac{f(n)}{n} e^{-it(\log x - \log n)} \right| dt \\ &\ll \int_{|t|\leq T} |G(t)||F(1+it)|\ dt.\end{aligned}$$

But from integration by parts we see that $G(t) \ll \frac{1}{1+|t|}$, and the claim follows. □

Next, we recall a standard L^2 mean value estimate for Dirichlet series:

> **Proposition 4 (L^2 mean value estimate)** Let $f : \mathbf{N} \to \mathbf{C}$ be an arithmetic function, and let $T \geq 1$. Assume that the Dirichlet series $F(s) := \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$ is absolutely convergent for $\mathrm{Re}(s) \geq 1$. Then
>
> $$\int_{|t|\leq T} |F(1+it)|^2\ dt \ll T \sum_{n_1,n_2 : \log n_2 = \log n_1 + O(1/T)} \frac{|f(n_1)||f(n_2)|}{n_1 n_2}.$$

Proof: This follows from Lemma 7.1 of Iwaniec-Kowalski; for the convenience of the reader we reproduce the short proof here. Introducing the normalised sinc function $\mathrm{sinc}(x) := \frac{\sin(\pi x)}{\pi x}$, we have

$$\begin{aligned}\int_{|t|\leq T} |F(1+it)|^2\ dt &\ll \int_{\mathbf{R}} |F(1+it)|^2 \mathrm{sinc}^2(t/2T)\ dt \\ &= \sum_{n_1,n_2} f(n_1)\overline{f(n_2)} \int_{\mathbf{R}} \frac{1}{n_1^{1+it} n_2^{1-it}} \mathrm{sinc}^2(t/2T)\ dt \\ &\ll \sum_{n_1,n_2} \frac{|f(n_1)||f(n_2)|}{n_1 n_2} \left| \int_{\mathbf{R}} \mathrm{sinc}^2(t/2T) e^{it(\log n_1 - \log n_2)}\ dt \right|.\end{aligned}$$

But a standard Fourier-analytic computation shows that $\int_{\mathbf{R}} \mathrm{sinc}^2(t/2T) e^{it(\log n_1 - \log n_2)}\, dt$ vanishes unless $\log n_2 = \log n_1 + O(1/T)$, in which case the integral is $O(T)$, and the claim follows. □

Now we recall a basic sieve estimate:

> **Proposition 5 (Sieve bound)** Let $x \geq 1$, let I be an interval of length x, and let $\mathcal{P}$ be a set of primes up to x. If we remove one residue class mod p from I for every $p \in \mathcal{P}$, the number of remaining natural numbers in I is at most $\ll |I| \prod_{p \in \mathcal{P}} (1 - \frac{1}{p})$.

Proof: This follows for instance from the fundamental lemma of sieve theory (see e.g. Corollary 19 of this blog post). (One can also use the Selberg sieve or the large sieve.) □

Finally, we record a standard estimate on the number of smooth numbers:

> **Proposition 6** Let $u \geq 1$ and $\varepsilon > 0$, and suppose that x is sufficiently large depending on u, ε. Then the number of natural numbers in $[1, x]$ which have no prime factor larger than $x^{1/u}$ is at most $O(u^{-(1-\varepsilon)u} x)$.

Proof: See Corollary 1.3 of this paper of Hildebrand and Tenenbaum. (The result also follows from the more classical work of Dickman.) We sketch a short proof here due to Kevin Ford. Let $\mathcal{S}$ denote the set of numbers that are "smooth" in the sense that they have no prime factor larger than $x^{1/u}$. It then suffices to prove the bound

$$\sum_{n \in \mathcal{S}: n \leq x} \log n \ll \exp(-u \log u + O(u)) x \log x, \tag{4}$$

since the contribution of those n less than (say) $\sqrt{x}$ is negligible, and for the other values of n, $\log n$ is comparable to $\log x$. Writing $\log n = \sum_{dm=n} \Lambda(d)$, we can rearrange the left-hand side as

$$\sum_{m \in \mathcal{S}: m \leq x} \sum_{d \in \mathcal{S}: d \leq x/m} \Lambda(d).$$

By the prime number theorem, the contribution to $\sum_{d \in \mathcal{S}: d \leq x/m} \Lambda(d)$ of those $d \leq x^{1/u}$ is $O(\min(x^{1/u}, x/m)$, and the contribution of those d with $d > x^{1/u}$ consists only of prime powers, which contribute $O((x/m)^{1/2})$. Combining these estimates, we can get a bound of the form

$$\sum_{d \in \mathcal{S}: d \leq x/m} \Lambda(d) \ll (x/m)^{1-\rho} (x^{1/u})^{\rho}$$

where $0 < \rho < 1/2$ is a quantity to be chosen later. Thus we can bound the left-hand side of (4) by

$$O(x^{\rho/u} x^{1-\rho} \sum_{m \in \mathcal{S}} \frac{1}{m^{1-\rho}})$$

which by Euler products can be bounded by

$$O(x^{\rho/u} x^{1-\rho} \exp(\sum_{p \leq x^{1/u}} \frac{1}{p^{1-\rho}})).$$

By the mean value theorem applied to the function $t \mapsto \exp(\rho t \log x^{1/u})$, we can bound $p^\rho = \exp(\rho \log p)$ by $1 + \frac{\log p}{\log x^{1/u}} \exp(\rho \log x^{1/u})$ for $p \leq x^{1/u}$. By Mertens' theorem, we thus get a bound of

$$O(x^{\rho/u} x^{1-\rho} \exp(\log\log x^{1/u} + O(\exp(\rho \log x^{1/u}))).$$

If we make the choice $\rho := \frac{\log u}{\log x^{1/u}}$, we obtain the required bound (4). □

— 2. Proof of theorem —

By increasing M as necessary we may assume that $M \geq 10$ (say). Let $0 < \varepsilon_2 < \varepsilon_1 \leq 1/2$ be small parameters (depending on M) to be optimised later; we assume x to be sufficiently large depending on $\varepsilon_1, \varepsilon_2, T$. Call a prime p *small* if $p \leq x^{\varepsilon_2}$, *medium* if $x^{\varepsilon_2} < p \leq x^{\varepsilon_1}$, and *large* if $x^{\varepsilon_1} < p \leq x$. Observe that for any $n \leq x$ we can factorise f as a Dirichlet convolution

$$f(n) = f_1 * f_2 * f_3(n)$$

where

- f_1 is the restriction of f to those natural numbers n whose prime factors are all small;
- f_2 is the restriction of f to those natural numbers n whose prime factors are all medium;
- f_3 is the restriction of f to those natural numbers n whose prime factors are all large.

We thus have

$$\frac{1}{x} \sum_{n \leq x} f(n) = \frac{1}{x} \sum_{n \leq x} f_1 * f_2 * f_3(n). \tag{5}$$

It is convenient to remove the Dirac function $\delta(n) := 1_{n=1}$ from f_2, f_3, so we write

$$f_2 = \delta + f_2'; \quad f_3 = \delta + f_3'$$

and split

$$f_1 * f_2 * f_3 = f_1 * f_2 + f_1 * f_3' + f_1 * f_2' * f_3'.$$

Note that $f_1 * f_2$ is the restriction of f to those numbers $n \leq x$ whose prime factors are all small or medium. By Proposition 6, the number of such n can certainly be bounded by $O(e^{-1/\varepsilon_1} x)$ if x is sufficiently large. Thus the contribution of this term to (5) is $O(e^{-1/\varepsilon_1})$.

Similarly, $f_1 * f_3'$ is the restriction of f to those numbers $n \leq x$ which contain at least one large prime factor, but no medium prime factors. By Proposition 5 the number of such n is bounded by $O(\frac{\varepsilon_2}{\varepsilon_1} x)$ if x is sufficiently large. Thus the contribution of this term to (5) is $O(\frac{\varepsilon_2}{\varepsilon_1})$, and hence

$$\frac{1}{x} \sum_{n \leq x} f(n) \ll \left| \frac{1}{x} \sum_{n \leq x} f_1 * f_2' * f_3'(n) \right| + e^{-1/\varepsilon_1} + \frac{\varepsilon_2}{\varepsilon_1}.$$

Note that $f_1 * f_2' * f_3'$ is only supported on numbers whose prime factors do not exceed x, so the Dirichlet series of $f_1 * f_2' * f_3'$ is absolutely convergent for $\mathrm{Re}(s) \geq 1$ and is equal to $F_1(s) F_2'(s) F_3'(s)$, where F_1, F_2', F_3' are the Dirichlet series of f_1, f_2', f_3' respectively. Since $f_1 * f_2' * f_3'$ is bounded in magnitude by 1 (being a restriction of f), we may apply Proposition 3 and conclude (for x large enough, and discarding the $\frac{1}{1+|t|}$ denominator) that

$$\frac{1}{x}\sum_{n\leq x} f(n) \ll \int_{|t|\leq T} |F_1(1+it)||F_2'(1+it)| \mid F_3'(1+it)|\, dt$$

$$+\frac{1}{T} + e^{-1/\varepsilon_1} + \frac{\varepsilon_2}{\varepsilon_1}.$$

We now record some L^2 estimates:

Lemma 7 For sufficiently large x, we have

$$\int_{|t|\leq T} |F_2'(1+it)|^2\, dt \ll \frac{\varepsilon_1}{\varepsilon_2^2 \log x}$$

and

$$\int_{|t|\leq T} |F_3'(1+it)|^2\, dt \ll \frac{1}{\varepsilon_1^2 \log x}.$$

Proof: We just prove the former inequality, as the latter is similar. By Proposition 4, we have

$$\int_{|t|\leq T} |F_2'(1+it)|^2\, dt \ll T \sum_{n_1,n_2:\, \log n_1 = \log n_2 + O(1/T)} \frac{|f_2'(n_1)||f_2'(n_2)|}{n_1 n_2}.$$

The term $f_2'(n_1)$ vanishes unless $n_1 \geq x^{\varepsilon_2}$, and we have $n_2 = (1+O(1/T))n_1$, so we can bound the right-hand side by

$$\ll T \sum_{n_1 \geq x^{\varepsilon_2}} \frac{|f_2'(n_1)|}{n_1^2} \sum_{n_2 = (1+O(1/T))n_1} \frac{|f_2'(n_2)|}{n}.$$

The inner summand is bounded by 1 and supported on those n_2 that are not divisible by any small primes. From Proposition 5 and Mertens' theorem we conclude that

$$\sum_{n_2 = (1+O(1/T))n_1} |f_2'(n_2)| \ll \frac{1}{T\varepsilon_2 \log x} n_1$$

and thus

$$\int_{|t|\leq T} |F_2'(1+it)|^2\, dt \ll \frac{1}{\varepsilon_2 \log x} \sum_{n_1} \frac{|f_2'(n_1)|}{n_1}$$

$$\ll \frac{1}{\varepsilon_2 \log x} \prod_{x^{\varepsilon_2} < p \leq x^{\varepsilon_1}} (1-\frac{1}{p})^{-1}$$

$$\ll \frac{\varepsilon_1}{\varepsilon_2^2 \log x}$$

as desired. □

We also have an L^∞ estimate:

Lemma 8 For sufficiently large x, we have

> $$|F_1(1+it)| \ll \frac{1}{\varepsilon_2} e^{-M} \log x$$
>
> for all $|t| \leq T$.

Proof: From Euler products, Mertens' theorem, and (1) we have

$$\begin{aligned} F_1(1+it) &= \prod_{p \leq x^{\varepsilon_2}} \sum_{j=0}^{\infty} \frac{f(p^j)}{p^{j(1+it)}} \\ &\ll \prod_{p \leq x^{\varepsilon_2}} \left| 1 + \frac{f(p) p^{-it}}{p} \right| \\ &\ll \exp\Big(\sum_{p \leq x^{\varepsilon_2}} \operatorname{Re} \frac{f(p) p^{-it}}{p} \Big) \\ &\ll \varepsilon_2 \log x \exp\Big(- \sum_{p \leq x^{\varepsilon_2}} \frac{1 - \operatorname{Re}\, f(p) p^{-it}}{p} \Big) \\ &\ll \varepsilon_2 \log x \exp\Big(-M + 2 \log \frac{1}{\varepsilon_2} \Big) \\ &\ll \frac{1}{\varepsilon_2} e^{-M} \log x \end{aligned}$$

as desired. ❐

Applying Hölder's inequality, we conclude that

$$\frac{1}{x} \sum_{n \leq x} f(n) \ll \frac{1}{\varepsilon_1^{1/2} \varepsilon_2^2} e^{-M} + e^{-1/\varepsilon_1} + \frac{\varepsilon_2}{\varepsilon_1}.$$

Setting $\varepsilon_1 := 1/M$ and $\varepsilon_2 := M^{1/6} e^{-M/3}$ we obtain the claim.

Comments (25)



normsinc · 24 November 2015

Is there a typo in the introduction of the normalised sinc function sinc(pi*x) = sin(pi*x) / (pi*x)? Should this be sinc(x) = sin(pi*x) / (pi*x) to match the wikipedia notation for normalised sinc?

[Corrected, thanks – T.]



Anonymous · 25 November 2015

There may be a typo in the inequality (1) in Theorem 1. As it is stated now, as soon as (1) holds for some M it holds for any larger M', and therefore the $(1+M)e^{-M}$ term could be made arbitrarily small in the conclusion of the theorem. Also, the fact that (1) requires a bound from above does not seem consistent with the corollary stated right after Theorem 1.

pybienvenu · 25 November 2015

Indeed, in (1) this should be $\geq M$.

[Corrected, thanks – T.]

John Mangual · 25 November 2015

Does Halasz inequality imply the prime number theorem?

Terence Tao · 25 November 2015

The prime number theorem is equivalent to the assertion that $\sum_{n \leq x} \mu(n) = o(x)$. Applying Halasz's inequality, we see that to prove the prime number theorem, one needs to obtain a non-trivial lower bound on $\mathrm{Re} \sum_p \frac{p^{-it}}{p}$ for any real t, which is equivalent to the non-vanishing of $\zeta(1+it)$. This is not surprising since the other proofs of the PNT already demonstrate the equivalence of the PNT with the lack of zeroes of zeta on the line $\mathrm{Re}\ s = 1$. Of course, one still has to establish the non-vanishing of $\zeta(1+it)$, and for this one can use any of the standard techniques, e.g. one can use the inequality $\mathrm{Re}\ 3 + 4p^{-it} + p^{-2it} \geq 0$ together with an upper bound for $\mathrm{Re} \sum_p \frac{p^{-2it}}{p}$ (or equivalently, an upper bound for $|\zeta(1+2it)|$).

In the pretentious approach to multiplicative number theory, Halasz's inequality is basically used as a replacement for complex methods (such as those based on Perron's formula), but does not directly address the most difficult part of the theory, which is to determine the location of the zeroes of various zeta and L-functions.

John Mangual · 25 November 2015

There were a few conceptual leaps in understanding the prime number theorem, if at all.

Even though $\sum \frac{1}{n}$ diverges, any twist is finite $0 < \sum \frac{1}{n^{1+it}} < \infty$. This is a very myopic – or astigmatic – way of looking at numbers; things are very blurry $x \approx 0, 1$ or ∞. All of these averages together prove the prime number theorem.

There are a few ways to show $\zeta(1+it) \neq 0$ in the book Titchmarsh. If I read them again, I might understand why that trig equality $3 + 4\cos\theta + \cos 2\theta \geq 0$ is so special here. IDK

I am also reminded of contest problems where we try to "paint" all real numbers which certain restrictions on rectangles, or triangles, or related pairs of points. Then the objective is to prove function must be 0.

What are mains is an "engine" which turns estimates of the zeta function into estimate about primes. Halasz inequality requires some multiplicative function with $|f(p)| \leq 1$ and somehow you estimate the averages in (1) and out pops this averaging result.

A priori if we add a bunch of complex numbers we can't assume the average is zero can we?

Anonymous · 25 November 2015

Do you think a different approach can be made in a less "pretentious" way?

Anonymous · 25 November 2015

Is condition (1) "natural" in some sense? (i.e. is it based on some motivation from which it "naturally" follows in more or less systematic way ?)

Terence Tao · 25 November 2015

One can view the quantity $(\sum_{p \leq x} \frac{1 - \mathrm{Re}\ f(p)\overline{g(p)}}{p})^{1/2}$ as a measure of "distance" between two multiplicative functions f, g; in the language of Granville and Soundarajan, this quantity measures the extent to which f "pretends" to be like g or vice versa. (With the square root, this distance obeys the triangle inequality.) The condition (1) is then asserting that the function $f(n)$ does not pretend to be like any of the Archimedean characters n^{it}.

Using the Euler product formula $\sum_n \frac{f(n)}{n^s} = \prod_p (1 + \frac{f(p)}{p^s} + \ldots)$, the condition (1) is also basically equivalent to an upper bound on the magnitude of the Dirichlet series of f on the line segment $\{1 + it : |t| \leq T\}$, which is already known (through the Perron formula) to be closely tied to the problem of obtaining asymptotics for $\frac{1}{x} \sum_{n \leq x} f(n)$.

Anonymous · 26 November 2015

This seems to motivate the following generalization of condition (1):
Let $M(x, \sigma, T)$ denote for $\sigma \leq 1$ the minimum of the LHS of (1) with its denominator generalized to p^σ.
This may give (for $M(x, \sigma, T) \geq 0$) generalized versions of theorems 1, 2.
Moreover, any additional information on possible meromorphic continuation of the Dirichlet series of f into the half plane $\sigma \leq 1$, may help to improve further the estimates.

jessemckeown · 26 November 2015

In the "cheap" version, is the algebraic term really meant to be 1/T ? for the interesting Ts, this is rather smaller than the "pretentious" version. (o.t.o.h, I suppose the interest is more for small M, where we get a lot for very little?)

Terence Tao · 26 November 2015

Yes, 1/T appears to be the natural dependence on T (as one also sees from the Granville-Soundararajan refinement of the original Halasz inequality, mentioned in the blog post); this type of error also shows up for instance in the truncated Perron formula (up to logarithmic losses, at least).

Bittu · 28 November 2015

Terry sir,I am a big fan of yours.I was proposing that calculus requires functions but most of the things around us are so irregular that a proper fitting function can't be given to it.Can you develop a superior language than calculus please.

[See https://en.wikipedia.org/wiki/Generalized_function -T.]

Anonymous · 29 November 2015

There seems to be a few typos still. Right after the heading "2. Proof of theorem" there is "By increasing M ias necessary..." Then about half a page below that "By Proopsition 5 th enumber..."

[Corrected, thanks – T.]

Kevin Ford · 1 December 2015

A short proof of Proposition 6 can be found here: http://www.math.uiuc.edu/~ford/sieve_notes_smoothnumbers.pdf

[Thanks for reminding me of your nice proof, I have added a version of it to the text. -T.]

> **Kevin Ford** · 2 December 2015
>
> I don't claim any originality in the proof. It just combines the method of Wirsing (logarithmic weights) with the "Rankin trick" (introducing the parameter ρ), both quite useful elementary tools for studying sums of multiplicative functions.

Anonymous · 1 December 2015

Dear Terry,
I'm always worry of you.I consider you like my brother.There's recently a man claims to solve Riemann hypothesis but not give any proof.I never believe him.Terry, even yourself,a great genius, also very hard to struggle with Riemann.I advice you should'nt devote your result of Riemann early.It's
dangerous .He certainly plagiarizes.This is a tip trick to trap you.At that time, he will have three things: money, fame, girls.The sixth sense tells me know about this.
Best regards,

Zak · 2 December 2015

I was curious which part of the above argument fails to achieve the better quantitative bounds, and what kind of ideas are needed for improvements.

> **Terence Tao** · 2 December 2015
>
> Adam Harper has an unpublished variant of this argument (from a few years back) which obtains the sharp bound. I think the main additional idea is to use the trick (which Kevin attributed to Wirsing in this comment thread) of summing $\sum_{n \leq x} f(n) \log n$ in place of $\sum_{n \leq x} f(n)$, and then decomposing $n \mapsto f(n) \log n$ as a triple convolution. (As far as I know, though, Adam has not yet made this argument publicly available.)

Mallesham · 17 February 2020

There is typo in the proof Lemma 8.

[Can you be more precise about this? – T.]

Mallesham · 21 February 2020

In the fourth and fifth lines of the proof of Lemma 8, I think it should be exponential of sum over primes instead of product over primes.

[Corrected, thanks – T.]

irvingxryang · 11 July 2021

for proof of Lemma 7, in the second line of expressions, you missed a term 1/n_2

[Corrected, thanks – T.]

Anonymous · 22 May 2024

I think the proof of Lemma 8 is probably wrong, though the result is right. In the second last inequality of your proof, the log log should be 2 loglog . If you drop these negative terms, which has at most a constant contribution in the initial Euler expansion, you would obtain the loglog as desired.



Terence Tao · 22 May 2024

Can you clarify? I don't see any log log terms in the proof of Lemma 8.

Anonymous · 5 August 2025

The conclusion of Lemma 8 is actually wrong. If we take $f(p) = -1$ and $t = 0$, then $F_1(i + it) \asymp \epsilon^{-1}(\log x)^{-1}$ but $e^{-M} \log x \sim (\log x)^{-1}$. Hence the correct result shall be $F_1(i + it) \ll e^{-M}\epsilon_2^{-1} \log x$ and the exponent in the final result is $1/3$ in place of $1/2$.

[Corrected, thanks – T.]